

ARCHITECTURE & SYSTEM SPECIFICATION

SK BIKE POINT

Two-Wheeler Workshop Management System

Master UML Use Case, Database Schema, Class Architecture, Sequence Flows, Process Flowchart, and Timeline

TECHNOLOGY STACK

Spring Boot 3 • React 18 • Vite •
Tailwind

TARGET ARCHITECTURE

RESTful APIs • JWT RBAC •
Relational DB

DOCUMENT TYPE

Architecture Specification

FIGURE 1.0 (MASTER)

UML USE CASE

System Use Case Model

Overall actor interactions across Guest, Customer, Mechanic, and Workshop Admin roles.

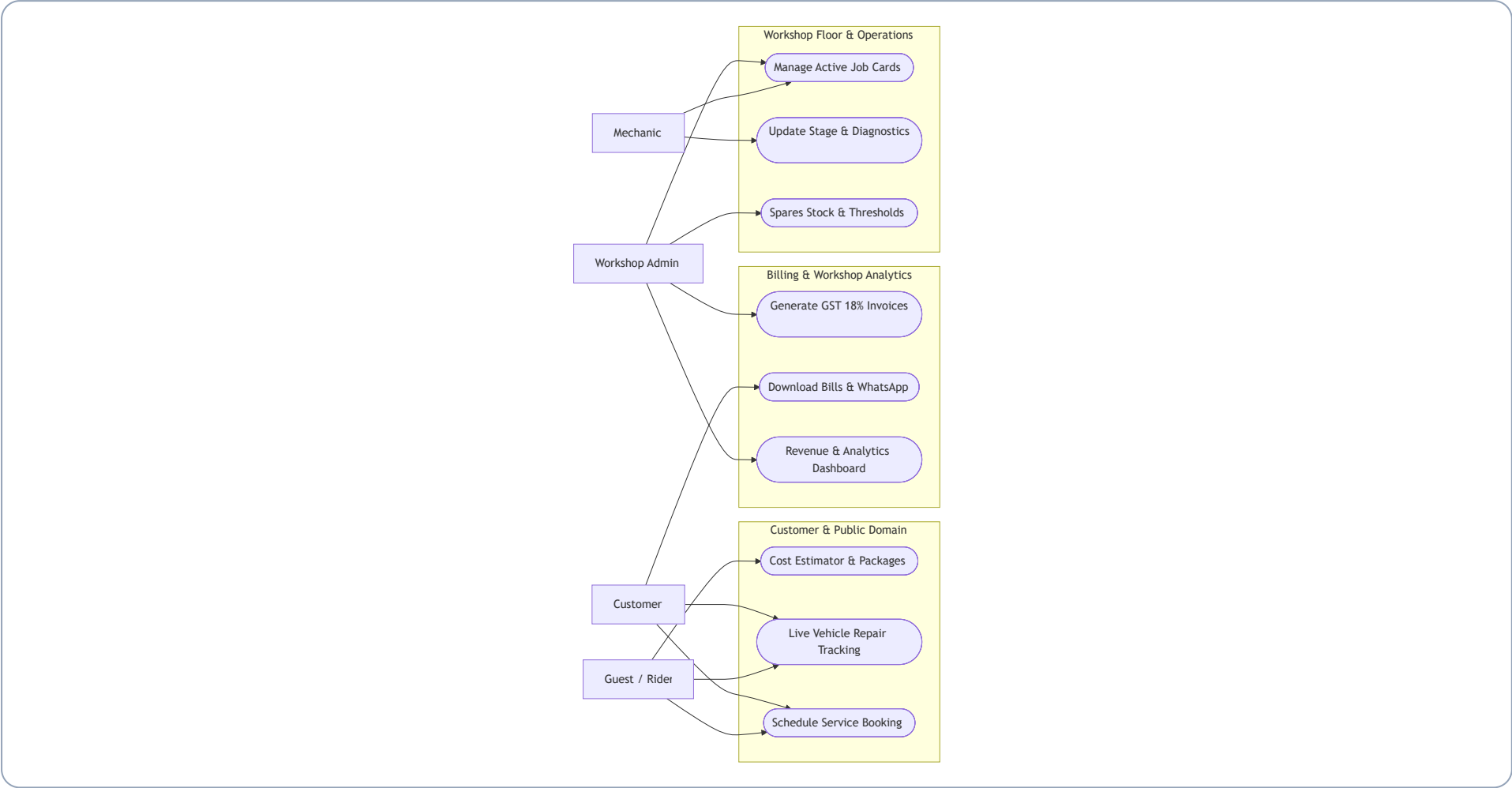


FIGURE 1.1 (SUB-DIAGRAM)

UML USE CASE

Workshop Floor Operations & Role Permissions

Floor job card management, mechanic allocation, inventory controls, and GST billing.

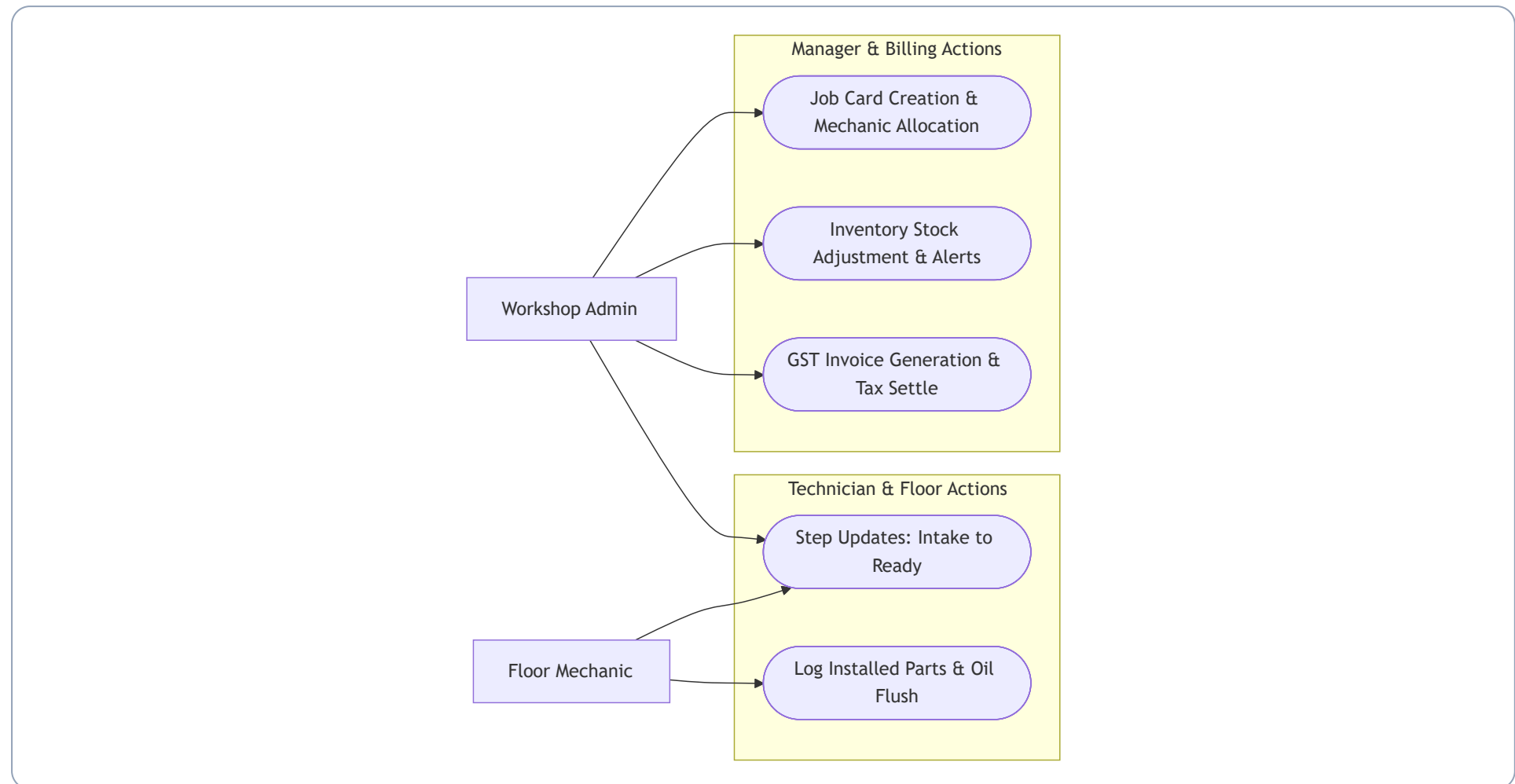


FIGURE 2.0 (MASTER)

DATABASE SCHEMA

Complete Entity-Relationship (ER) Schema

Complete database entity models, foreign keys, cardinality, and constraints.

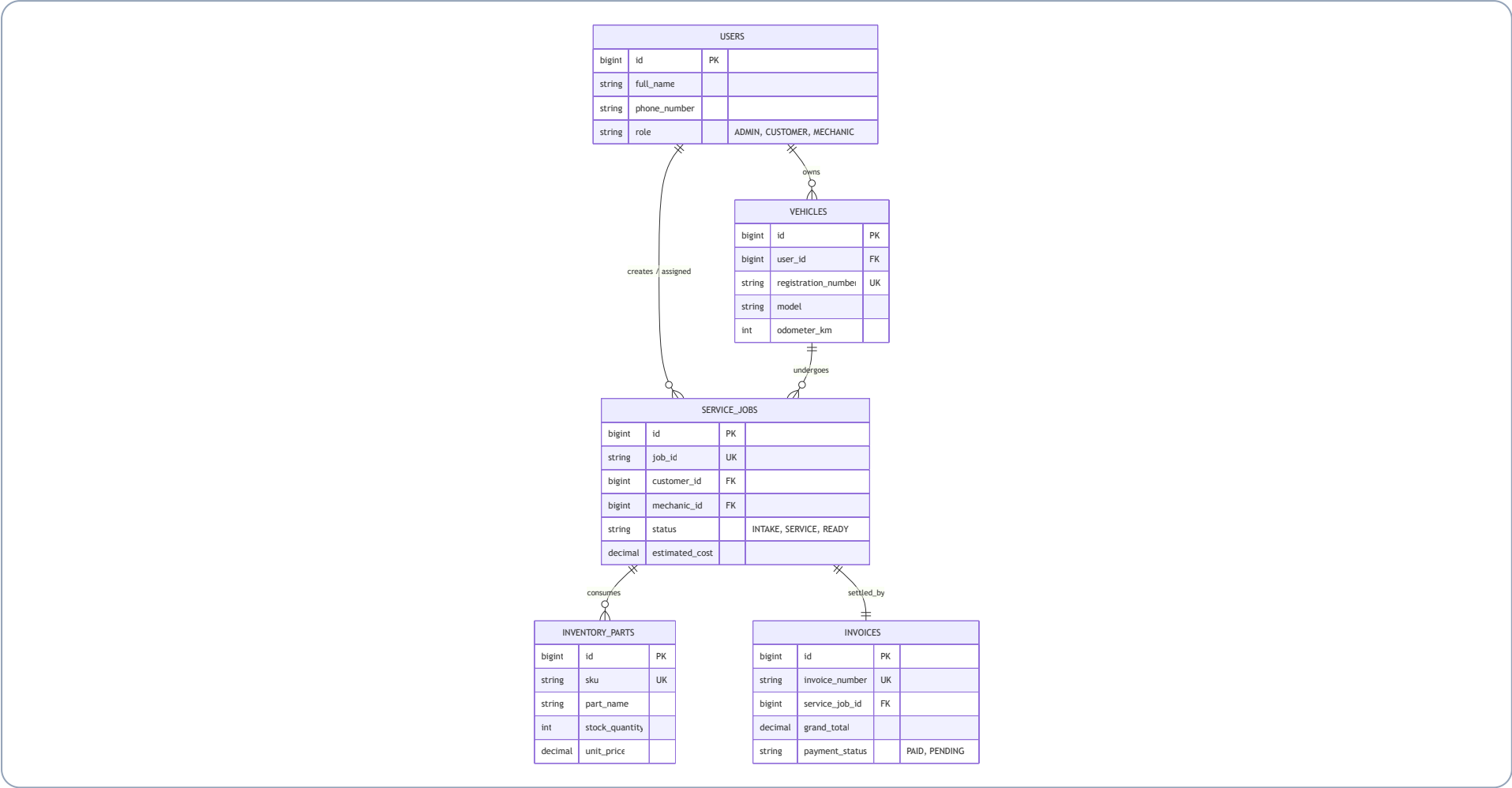


FIGURE 2.1 (SUB-DIAGRAM)

DATABASE SCHEMA

ER Model: Service Jobs, Spares & Invoicing

Job stage milestones, parts consumption mapping, stock ledger, and GST billing tables.

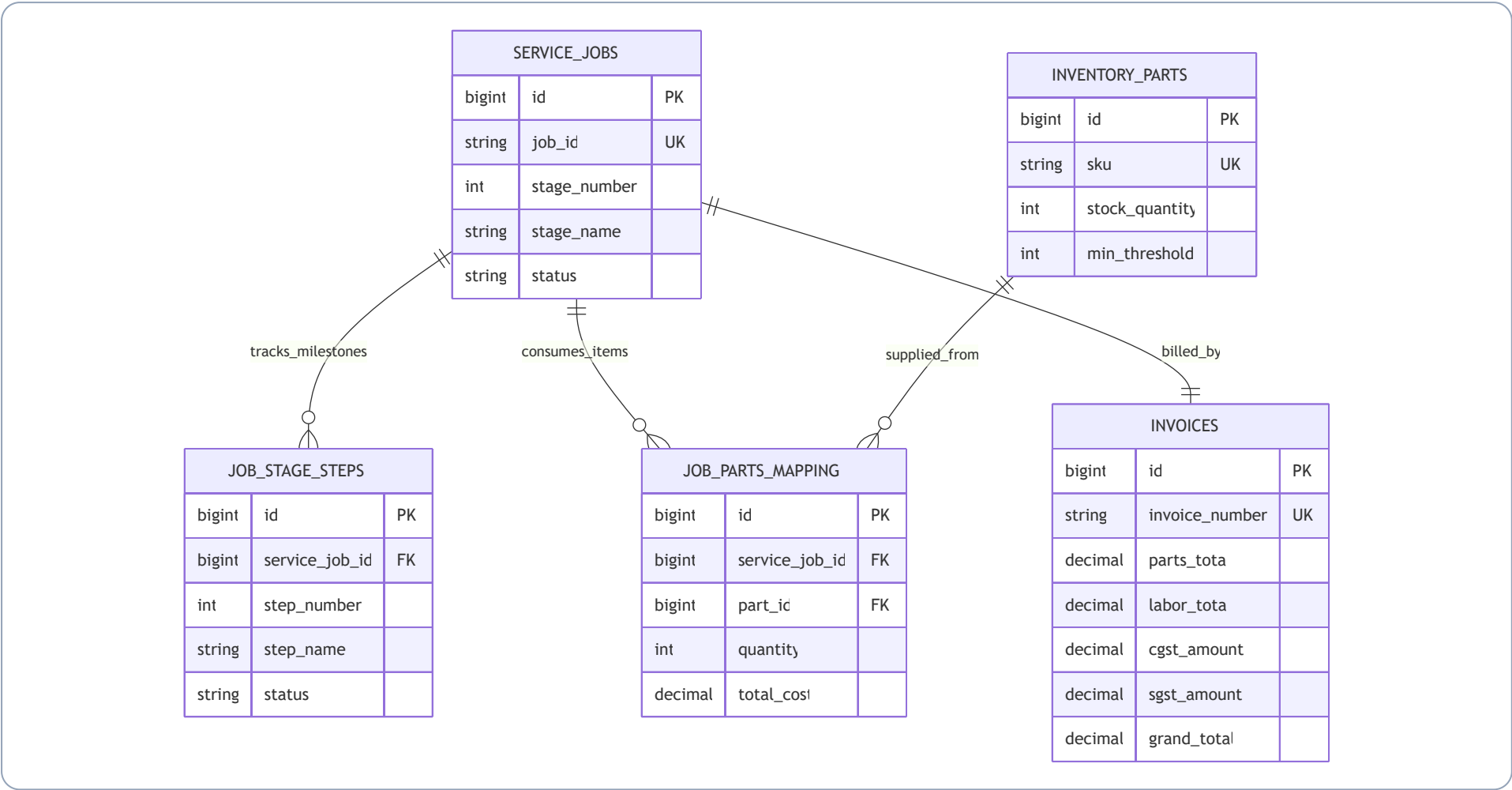


FIGURE 3.0 (MASTER)

UML CLASS MODEL

System Class Diagram

Spring Boot REST Controllers, Service interfaces, JPA Entities, and Data Transfer Objects.

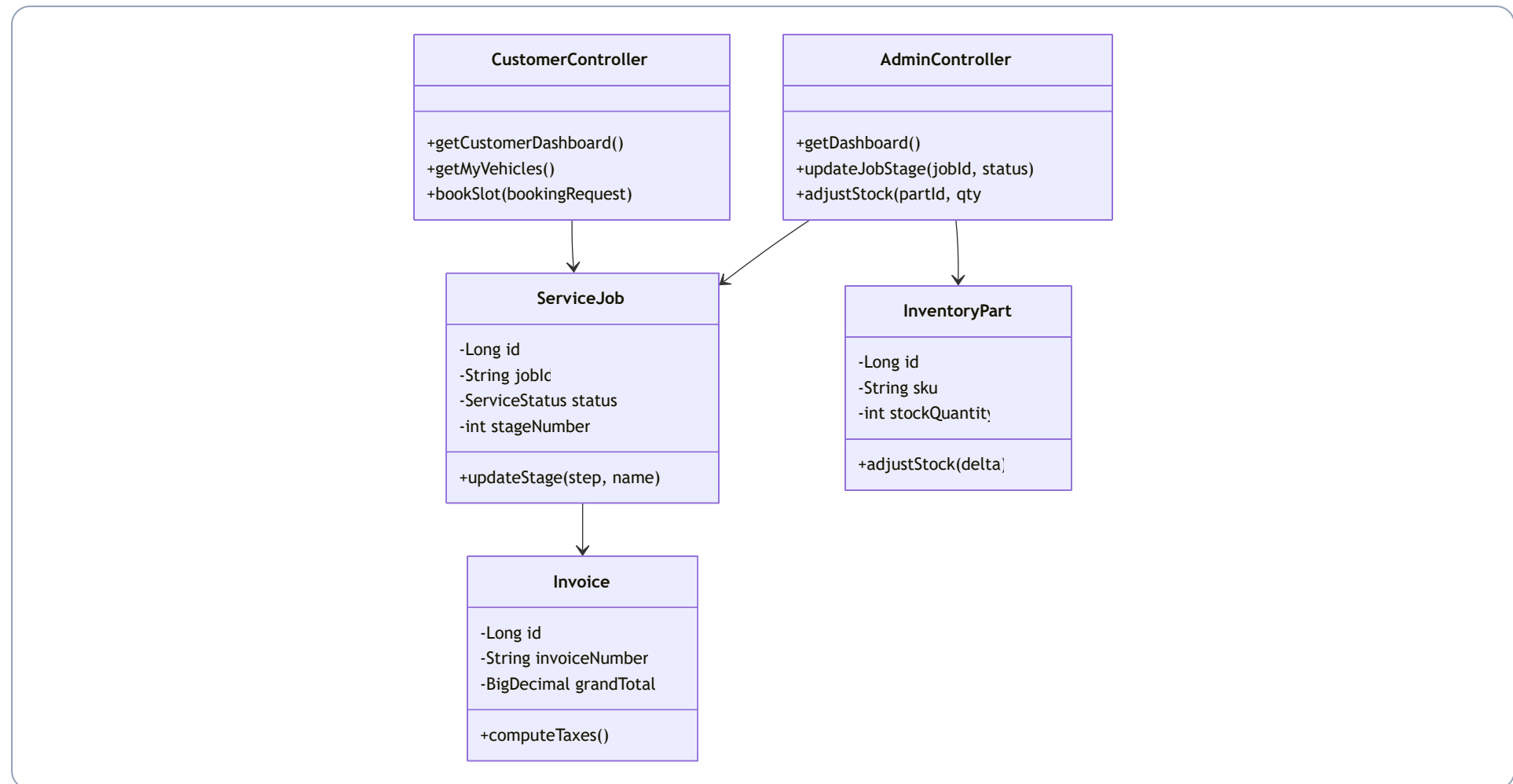


FIGURE 4.1 (SEQUENCE A)

UML SEQUENCE

Sequence: Live Repair Tracker & Booking Flow

Synchronous API query lifecycle for public vehicle status lookup and appointment scheduling.

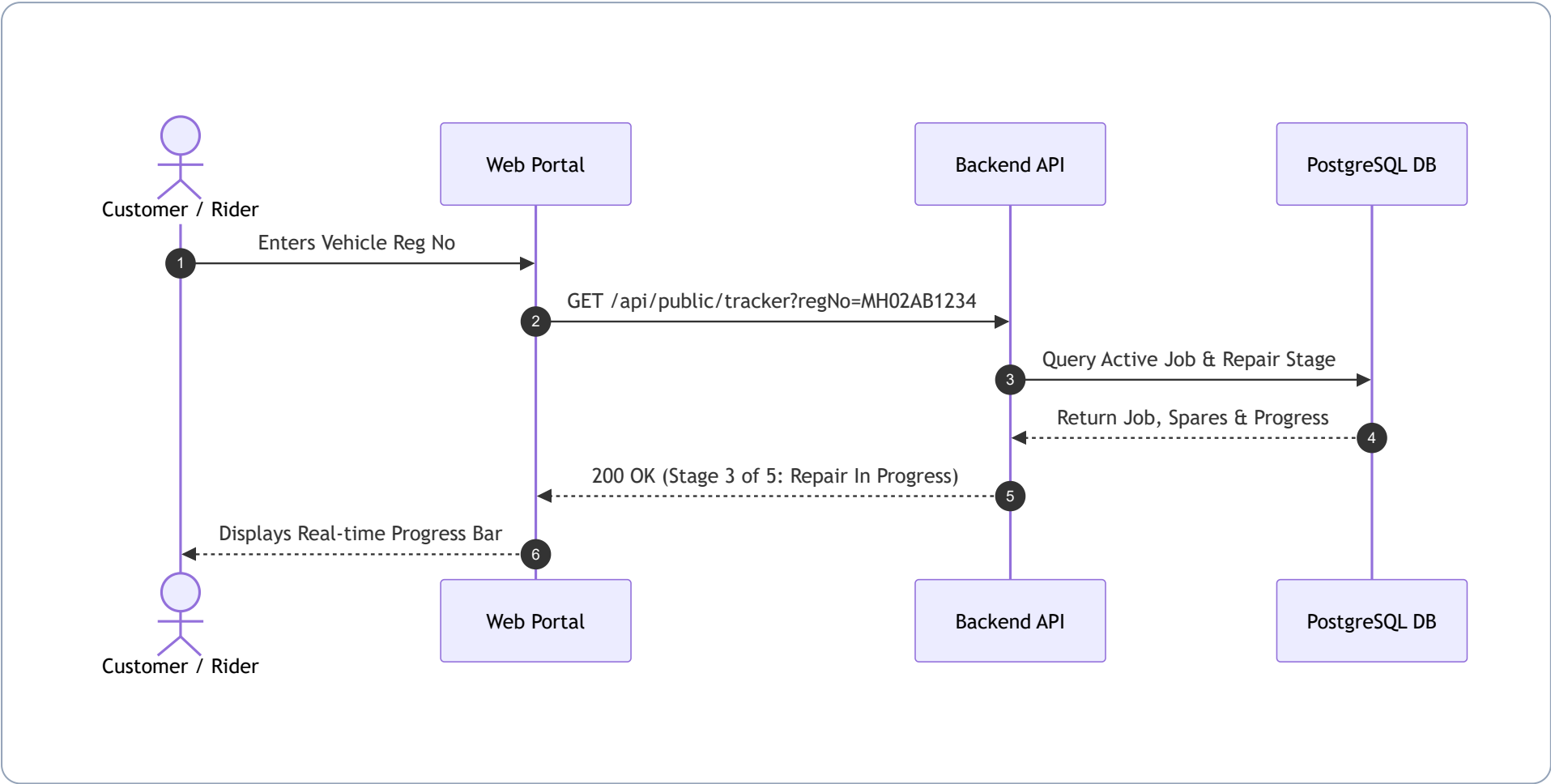


FIGURE 4.2 (SEQUENCE B)

UML SEQUENCE

Sequence: Floor Repair & GST Invoicing Flow

Workshop stage updates, inventory stock decrement, and GST 18% tax calculation workflow.

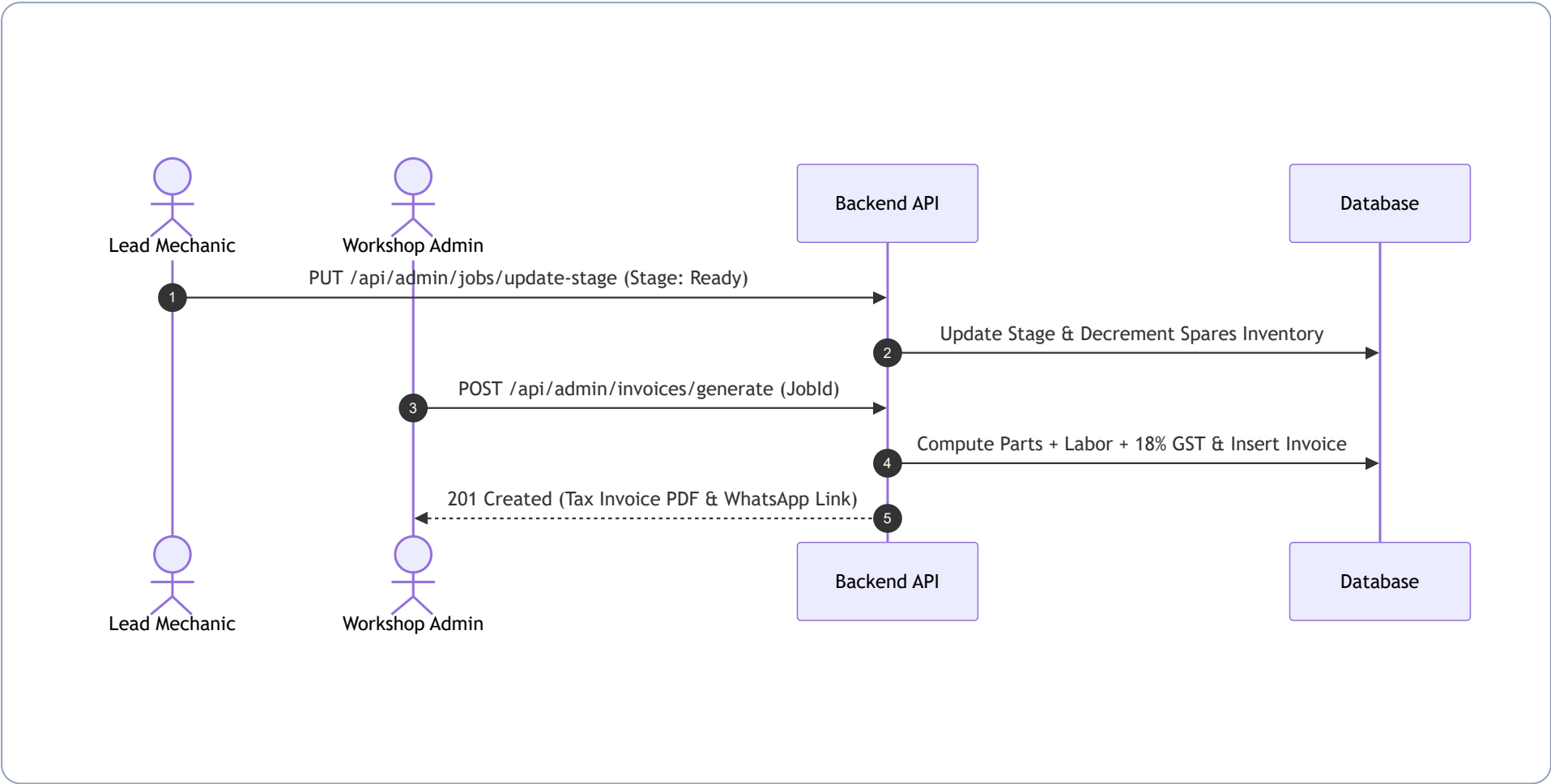


FIGURE 5.0 (MASTER)

PROCESS FLOWCHART

End-to-End Workshop Process Flowchart

Complete vehicle servicing lifecycle from intake to inspection, repair, wash, and gate pass handover.

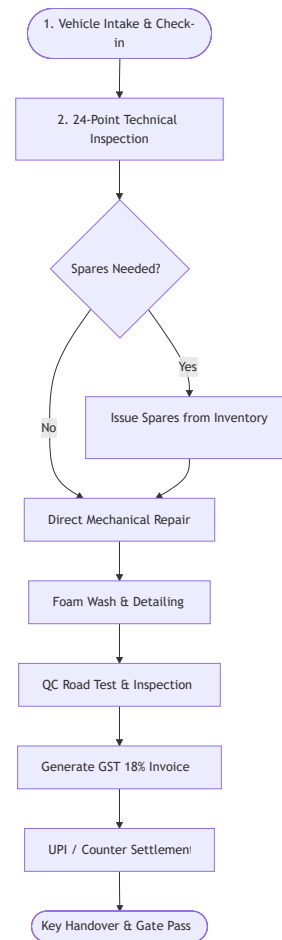


FIGURE 6.0 (MASTER)

PROJECT ROADMAP

Engineering & Implementation Roadmap

Gantt schedule covering system requirements, architecture, module development, and verification.

